

Examples

1. The “Monty Hall” problem, revisited

You’re on a game show, and are presented with three doors. One hides a car; the other two hide goats. You’re allowed to choose a door. Monty Hall then opens one of the other two doors to reveal a goat. You are now allowed to either stick with the door you chose initially, or switch to the other closed door. What should you do (assuming that you are hoping to get the car, and not the goat): stick with your original choice, or switch?

(a) In the case that Monty never opens the door with the car.

(b) In the case that Monty is choosing at random between the 2 doors that you didn’t choose.

2. Mendel, revisited.

Mendel's peas had either purple or white flowers; flower color is due to a single gene, for which the purple allele (A) is dominant to the white allele (a).

We cross two pure-breeding lines (one purple and one white) to produce the F_1 hybrid. We self the F_1 and choose an F_2 seed at random. We grow and self the F_2 and choose two F_3 seeds at random.

Consider the following events.

- $P = \{F_2 \text{ has purple flowers}\}$
- $O = \{F_2 \text{ is homozygous } AA\}$
- $E = \{F_2 \text{ is heterozygous } Aa\}$
- $W = \{F_2 \text{ has white flowers}\}$
- $A_1 = \{F_3 \text{ number 1 has purple flowers}\}$
- $A_2 = \{F_3 \text{ number 2 has purple flowers}\}$

(a) Are A_1 and A_2 independent?

(b) Calculate $\Pr(A_1 \text{ and } A_2 \mid E)$.

(c) Calculate $\Pr(A_1 \text{ and } A_2 \mid \mathbf{P})$.

(d) Calculate $\Pr(A_1 \text{ and } A_2)$.

3. Suppose the test for HIV correctly gives a positive result, if a person is infected, with probability 99.5%, and correctly gives a negative result, if a person is not infected, with probability 98%.

(a) Suppose that 0.1% of a population are infected with HIV. Consider drawing a person at random and testing them for HIV infection.

Calculate $\Pr(\text{infected} \mid \text{test is positive})$.

(b) Consider a person drawn from a very high-risk group, so that they have, *a priori*, probability 30% of being infected.

Calculate $\Pr(\text{infected} \mid \text{test is positive})$.